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ARTICLE

Applications of AI-based deep learning models for detecting dental caries on intraoral images – a systematic review

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KEY POINTS

- Artificial intelligence (AI) is a rapidly growing disruptive technology with various applications in dentistry.
- Different deep learning models have been developed to detect dental caries using intraoral images.
- Levels of deployment and explainable AI were important parameters which are reported in a few studies.

OBJECTIVES: This systematic review aimed to assess the effectiveness of Artificial Intelligence (AI)-based Deep Learning (DL) models in the detection of dental caries on intraoral images.

METHODS: This systematic review adhered to PRISMA 2020 guidelines conducting an electronic search on PubMed, Scopus, and CENTRAL databases for retrospective, prospective, and cross-sectional studies published till 1st June 2024. Methodological and performance metrics of clinical studies utilizing DL models were assessed. A modified QUADAS risk of bias tool was used for quality assessment.

RESULTS: Out of 273 studies identified, a total of 23 were included with 19 studies having a low risk and 4 studies having a high risk of bias. Overall accuracy ranged from 56% to 99.1%, sensitivity ranged from 23% to 98% and specificity ranged from 65.7% to 100%. Only 3 studies utilized explainable AI (XAI) techniques for caries detection. A total of 4 studies exhibited a level 4 deployment status by developing mobile or web-based applications.

CONCLUSION: AI-based DL models have demonstrated promising prospects in enhancing the detection of dental caries, especially in terms of low-resource settings. However, there is a need for future deployed studies to enhance the AI models to improve their real-world applications.

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INTRODUCTION

Dental caries is the most common oral disease affecting more than 2 billion adults and 520 million children globally[1]. Despite the associated morbidities, dental caries detection and treatment remain an underfunded aspect of public health. Dental caries continues to be a significant challenge in Low-Middle-Income Countries (LMICs), primarily in under-resourced settings[2,3]. Prompt detection of dental caries can significantly reduce the overall disease burden, eventually reducing the cost and time associated with treatment.

Conventionally, dental caries is detected using visual-tactile and radiographic examinations. However, clinical detection of dental caries is largely based on the subjective judgment of clinicians which in turn is based on years of experience[4]. Furthermore, previous studies suggest that the combination of clinical and radiographic examination for detecting carious lesions improves sensitivity (39% to 94%)[5].

Artificial Intelligence (AI) is a rapidly evolving disruptive technology that has found numerous applications in healthcare. AI-based models have demonstrated a higher accuracy and sensitivity in the detection of various disorders such as skin lesions, pulmonary nodes, and different types of cancers[6,7]. Hence with its expanding nature, AI applications have inevitably penetrated dentistry, including the detection of dental caries[8–11].

AI models have been proposed to assist in diagnosing dental caries through the analysis of intraoral images. These models include Machine Learning (ML) approaches like Support Vector Machines (SVM) and decision trees, as well as Deep Learning (DL) models such as Convolutional Neural Networks (CNNs), including RCNN and ResNet. DL allows for automated feature extraction making it more effective for image analysis and thus outperform ML models. As a result, CNNs have proven to be a reliable AI algorithm for this purpose[12]. A recent study reported a mean Average Precision (mAP) of 0.769 for caries detection using intraoral images with the DL CNN model[13]. Another study, similarly reported a high accuracy of 92.5% for the same.

AI-based models may augment the role of dentists in detecting dental caries by identifying them clinically in real-time during examination, especially in areas with limited access to dental care[14]. Dental facilities and providers are often unevenly distributed geographically, significantly impacting marginalized populations[2]. In LMICs, attracting dentists to underserved areas is particularly challenging due to economic constraints, cultural barriers, and the resource-intensive nature of establishing and maintaining dental practices[15].

A recent systematic review by Moharrami et al. evaluated the outcome measures of the AI-based models both traditional ML and DL, for the detection of dental caries on intraoral pictures[16].

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Table 1. Search strategy for Retrieval of Studies

Database	Search strategy	Results
PubMed	(Intraoral photographs OR Intraoral pictures OR Intraoral images OR Teeth images OR Teeth pictures OR Tooth pictures) AND (Artificial Intelligence based model OR AI based Algorithm OR AI-based deep learning model OR Convolutional Neural network based model OR AI based CNN model) AND (Artificial Intelligence based model OR AI based Algorithm OR AI based deep learning model OR Convolutional Neural network based model OR AI based CNN model) AND (Clinical Examinations OR Dental Examinations OR Oral Examinations [MeSH] OR Image annotations of experienced dentists OR Physical examination) AND (Sensitivity [MeSH] OR Specificity [MeSH] OR Precision OR F1 score OR Accuracy OR Recall OR Mean Average Precision OR mAP)	122
Scopus	("Intraoral photographs" OR "Intraoral pictures" OR "Intraoral images" OR "Teeth images" OR "Teeth pictures" OR "Tooth pictures") AND ("Artificial Intelligence" OR "Algorithm" OR "deep learning" OR "Neural network" OR "CNN model")	149
CENTRAL	(Intraoral photographs OR Intraoral pictures OR Intraoral images OR Teeth images OR Teeth pictures OR Tooth pictures) AND (Artificial Intelligence based model OR AI based Algorithm OR AI-based deep learning model OR Convolutional Neural network based model OR AI based CNN model) AND (Artificial Intelligence based model OR AI based Algorithm OR AI based deep learning model OR Convolutional Neural network based model OR AI based CNN model) AND (Clinical Examinations OR Dental Examinations OR Oral Examinations OR Image annotations of experienced dentists OR Physical examination) AND (Sensitivity OR Specificity OR Precision OR F1 score OR Accuracy OR Recall OR Mean Average Precision OR mAP)	2

Although the authors had an extensive search strategy, there were certain shortcomings. The literature search for this review was carried out till December 2022, despite being published in 2024. Therefore, current relevant studies for the two years leading up to the publication process need to be included[13,17–19]. Various aspects of the studies were adequately explained, however, important aspects such as explainable AI (XAI) or the deployment of AI models for caries detection were not addressed. Additionally, the comparison of AI models with dentists was also not taken into account. In this review, traditional ML models were excluded, reflecting the growing emphasis in recent research on DL techniques. Traditional ML models often encounter limitations, such as challenges with feature extraction and contextual understanding[20]. Therefore, to understand this landscape better and to ensure a complete and current understanding of AI's role in caries detection a comprehensive systematic review was conducted.

METHODS

This systematic review was conducted in concordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analysis (PRISMA) guidelines 2020 (PROSPERO ID: CRD42024557422)[21]. The research question was developed using Participants, Intervention, Comparison, and Outcome (PICO) criteria.

PICO

Research question. How effective are AI-based DL models for the detection of dental caries on intraoral images?

Participants. Intraoral images of the oral cavity focusing on teeth.

Interventions. DL models for the detection of dental caries.

Comparisons. Reference standards such as clinical dental examinations or image annotations by dentists.

Outcomes. Performance metrics of the AI model.

Eligibility criteria

Inclusion criteria:

1. Original peer-reviewed studies published in the English language.
2. Studies reporting the application of AI and its comparison with the dental experts in the detection of dental caries on intraoral images.
3. Studies reporting the performance of the AI algorithm in the form of outcome metrics.
4. In-vivo studies reporting diagnosis of caries using DL models.

Exclusion criteria:

1. Studies conducted on animals.
2. Studies based on a dataset consisting of panoramic radiographs, bitewing radiographs, and periapical radiographs.
3. Studies based on ML models excluding DL for caries detection.
4. Reviews, conference proceedings, letter to the editors and commentaries.

Search strategy

The authors (A.N, S.A.A), in collaboration with medical information specialists from The Aga Khan University Hospital, Pakistan developed a search strategy focusing on DL algorithms, intraoral images, and related outcome measures. Electronic databases such as PubMed, Scopus, and Cochrane Central Register of Controlled Trials (CENTRAL) were used to retrieve studies that met the research question. The search was conducted till 1st June 2024, based on MeSH terms and keywords. Other than the electronic databases, studies were also searched manually using Google Scholar and snowball referencing of published papers. The complete search strategy is presented in Table 1.

Selection of studies

A 3-pronged approach was used for screening titles, abstracts, and full-text review independently conducted by two authors (A.N, S.A.A). The initial list of articles was imported into EndNote version 21.0 (Clarivate Analysis US LLC) with the removal of duplicates. Full texts of potential studies were obtained and assessed as per the eligibility criteria. Any conflicts regarding inclusion and exclusion of the studies were resolved by the third author (A.L).

Data extraction

A predesigned data extraction form was drafted using Google Sheets (Google©), with the following variables: Study title, link, author, year of publication, population, dataset, picture capturing device, image pre-processing, range of teeth (single tooth/multiple teeth), train/validation/test (ratio or number), frequency of caries, reference standard, ground truth for diagnosis, image annotation tool, type of dentition (primary or permanent), teeth location (anterior or posterior), AI model, framed approach (classification or segmentation or detection), performance metrics Accuracy, Sensitivity/Recall, Specificity, F-1 score, Precision, Positive Predictive Value, Area Under Curve (AUC), Receiver Operating Characteristics (ROC), Intersection Over Union (IOU) threshold, mAP/Average Precision (AP), XAI, level of deployment and comparison with dentists. Information based on these variables was extracted from the papers that met the eligibility criteria.

Risk of bias assessment

Quality assessment of the included papers was conducted by two independent authors using a modified Quality Assessment of

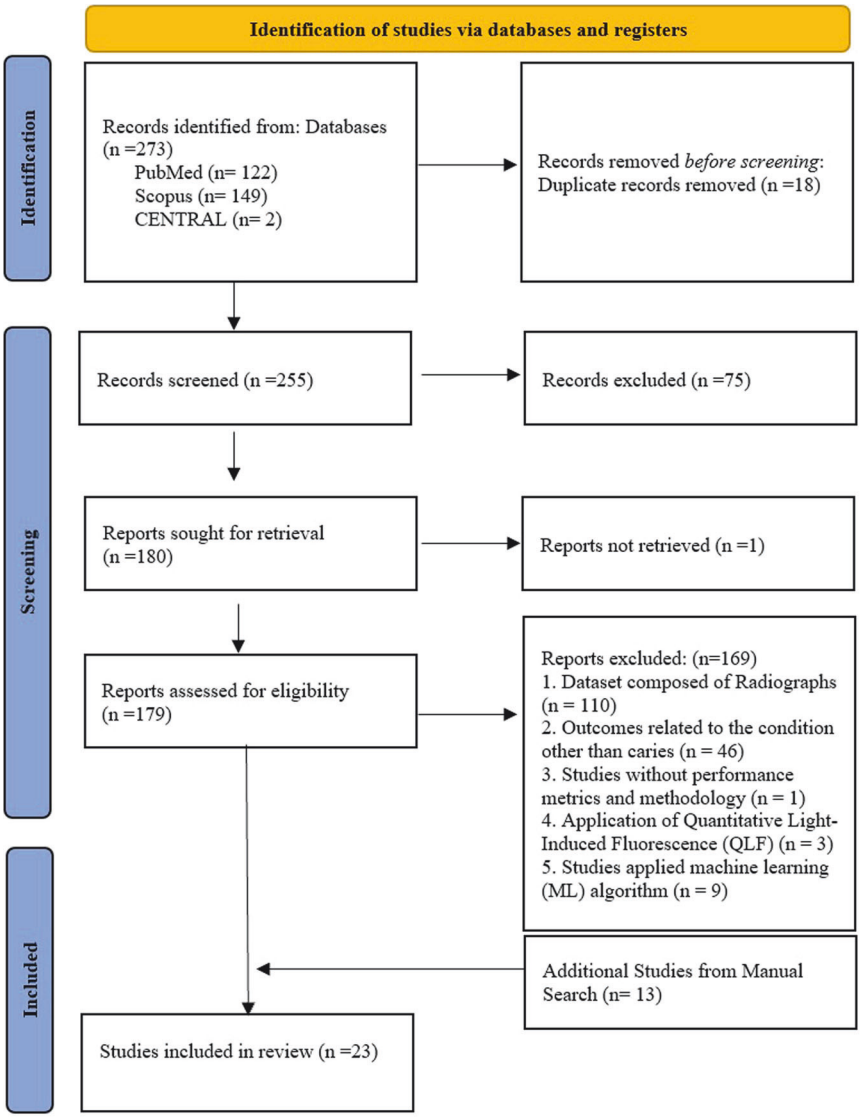


Fig. 1 PRISMA flowchart for the search process.

Diagnostic Accuracy Studies (QUADAS-2) tool[2,22]. The quality assessment questions pertained to performance metrics, information for future research replicability, use of validation techniques, and reference standards. The final methodological quality of the included studies was rated according to the following scale: score of 6–8 (high quality), score of 4–5 (moderate quality), and score of 0–3 (low quality) by the authors. Any difference of opinion and variance among the authors was resolved by the third author with discussion.

RESULTS

Study selection and search results

The initial search yielded a total of 273 records out of which 18 were excluded as duplicates. After screening titles of 255 papers, 75 papers were excluded on the basis of irrelevancy. A total of 180 papers underwent abstract screening out of which 1 paper could not be retrieved despite emailing the corresponding author to access the full text of the paper. Of the remaining 179 studies, 169 were excluded which did not fulfill the eligibility criteria hence 10 studies were included after the full-text reading stage. The studies identified using other methods such as manual search yielded another 13 studies which were included. A total of 23 studies were included in this review as illustrated in Fig. 1.

Evaluation of risk of bias

Modified QUADAS-2 tool was used in this systematic review to evaluate the risk of bias in the selected studies as presented in Table 2 **and** Fig. 2[3]. Out of all 23 studies, only 4 studies have shown a high risk of bias in 6 domains of the modified QUADAS-2 tool[17,23–25]. A sample size of over 100 images was used by 19 studies that assessed the AI model based on two or more performance parameters[13,17–19,23,25–38]. A low risk of bias was reported by 11 articles because they applied validation methodologies and provided sufficient information for the study to be replicated[13,18,26–31,33,37,39]. Only 9 out of 23 studies explicitly mentioned their inclusion and exclusion criteria for the retrieval of intraoral images[13,19,26,27,29,31,35,40,41].

STUDY CHARACTERISTICS

The characteristics of the included studies are summarized chronologically in SI Table 1. The year of publication of studies ranged from 2019 to 2024. Most of the studies were conducted in the Asian continent, mainly China (n = 8) and Korea (n = 3) however, a few studies were conducted in European and North American continents. Most of the data was retrieved from hospitals and healthcare facilities, although 5 studies used web scraping and internet archives to acquire images, with Kaggle© being the primary repository in 4 studies[17,23,24,29]. Overall, the sample size of the 23 included studies ranged from 74 to 24578 images, with 16 studies reporting a sample size equal to or more than 1000 images[13,18,19,25–28,30–38]. However, none of the studies reported methods used for sample size calculation.

Out of 23 studies, 11 focused on single tooth images for caries detection, while the rest used images comprising of multiple

Table 2. Quality Assessment of Included Studies Using Modified-QUADAS tool demonstrating the overall scores of each study reflecting its quality.

Risk of Bias Assessment	Does the sample size consist of >100 images?	Was there an explicit inclusion/ exclusion criterion defined for dataset collection?	Did the investigators mention sample size calculation?	Does the dataset involve multi-centers?	Was there adequate information provided for replicating study?	Did the investigators employ any validation techniques in the test data to reduce bias?	Does the study involve more than one specialist for annotations?	Did the investigators report more than one performance metrics?	Overall Score
Askar et al.	Low	Low	High	High	Low	Low	Low	Low	6
Ding et al.	Low	Low	High	Low	Low	Low	High	Low	6
Yu et al.	Low	High	High	Low	Low	High	Low	Low	5
Jiang et al.	Low	Unclear	High	Unclear	Low	Low	Low	Low	5
Kim et al.	Low	Unclear	High	High	Low	Low	Low	Low	5
Kuhnisch et al.	Low	Low	High	Unclear	Low	Low	High	Low	5
Moutselos et al.	High	Unclear	High	Low	Low	Low	Low	Low	5
Park et al.	Low	Unclear	High	High	Low	Low	High	Low	4
Sonavane et al.	High	High	High	Unclear	Unclear	Low	Unclear	High	1
Thanh et al.	Low	Unclear	High	Low	Low	High	High	Low	4
Zhang et al. 2020	Low	High	High	Unclear	Unclear	Low	Low	Low	4
Zhang et al. 2021	Low	High	High	Unclear	Unclear	Low	High	High	2
Javid et al.	High	Low	High	Low	Low	High	High	Low	4
Wang et al.	Low	Low	High	Low	Low	High	Low	Low	6
Liu et al.	Low	Unclear	High	Low	Low	High	Low	Low	5
Saini et al.	Low	High	High	Low	High	Low	High	High	3
Khan et al.	Low	Low	High	Unclear	Low	Low	High	Low	5
Rashid et al.	High	Low	High	High	Low	Unclear	Low	Low	4
Zhang et al. 2022	Low	High	High	Low	Low	Low	Low	Low	6
Siji Rani	Low	High	High	High	High	Low	High	High	2
Xiong et al.	Low	High	High	High	Low	Low	Low	Low	5
M. Mehdizadeh et al.	Low	Low	High	Low	High	Low	Unclear	Low	5
Yoon et al.	Low	Low	High	High	Low	Low	Low	Low	6

Low: 0–3 scores, Moderate: 4-5 scores, High: 6–8 scores.

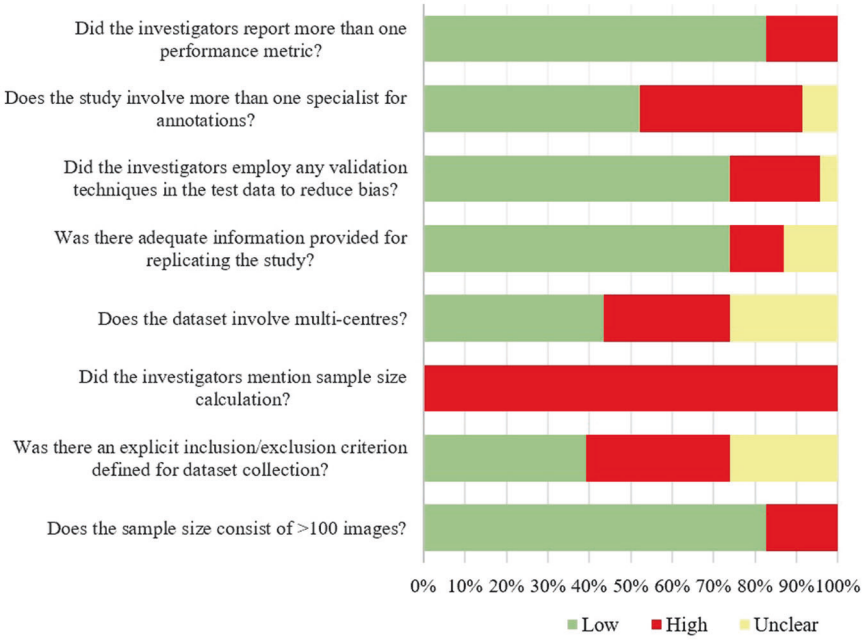


Fig. 2 The proportion of studies evaluated on various criteria with low, high, and unclear risk of bias.

teeth[17,23–26,30,31,33,35,36,39]. The prevalence of carious lesions was described by 11 studies with significant variance amongst the dataset of individual studies[17–19,26,28,30–34,37]. Few studies classified the carious lesions as follows: mild, moderate, and severe[28]; surface involvement as occlusal and proximal, of non-cavitated lesions as well as cavitated ones[31].

Modified International Caries Detection and Assessment System (ICDAS) was used as a reference standard by 4 studies[31,33,38,39] while 1 study used the International Caries Classification and Management System (ICCMS)[34].

A total of 18 studies utilized both anterior and posterior teeth images for training and testing of their AI models[18,27–34,37,38,41]. Only 3 studies focused on posterior teeth[23,36,39] and 2 studies focused exclusively on anterior teeth[26,40].

The performance metrics of the included studies are summarized in SI Table 2. Different image pre-processing techniques had been applied in the selected studies; to resize, enrich, refine, and augment images, before employing them to the AI models, except for 2 studies[25,39]. Some studies utilized image annotation softwares such as LabelImg ©, Labelme ©, Labelstudio ©, and Dental Annotator for identifying carious lesions.

However, the majority of the studies did not mention the annotation tool employed. Additionally, one or more experienced dentists annotated the dataset, in 14 studies, to provide ground truth for diagnosis and model training[13,18,19,26,30–37,40,41]. Although most of the studies did not mention the training process of annotators to ensure accuracy and consistency of labelling. Overall, the accuracy, sensitivity, and specificity of each study is illustrated in Fig. 3.

Stratification of carious lesions with task

A total of 2 studies performed classification and segmentation with the approach of detecting white spot lesions with an accuracy of 95.50% and 84%, respectively[26,35]. A single study utilized the ICDAS II classification system (classes 0–6) to segment exclusively permanent molars with an F1 score of 59.5, 64.2, 37.7, 60.0, 47.8, 71.9, and 87, respectively[39]. Only 1 study employed both the ICDAS and the American Dental Association Caries Classification System (ADA CCS) for caries detection and localization, reporting accuracies of 93.4%, 93.9%, and 97.7% for stages 1–3, respectively[13]. Detection of primary and secondary caries was done by 1 study on intraoral images which reported F1 scores of 80.0 and 69.0, respectively[27].

Inter-model comparisons

For inter-model comparisons, 8 out of 23 studies evaluated each model's accuracy with their datasets[17,19,23,25,28,30,34,36]. Out of 23 studies, 5 developed their own customized models, including DentalNet and UCDA, which are based on CNNs[18,26,29,35,36]. These models achieved accuracies of 94.25% and 95.25%, respectively. Training and comparison of pre-developed AI models such as ResNet-50, Vgg-16, Yolov3, RetinaNet, Faster-RCNN & GoogLeNet was carried out by 9 studies[17,19,23,25,28,30,34,36,38]. However, customized models were reported to have higher accuracies.

A study developed a customized model, RDFNet, which featured a transformer mechanism and was based on the Yolov5 backbone network[28]. It was compared with YOLOv5s, YOLOv5m, YOLOv5l, YOLOv3-tiny, Faster R-CNN, and Mask R-CNN. The accuracy of the RDFNet was 62.3% which was higher than the other models.

Comparing AI models with dental expertize

The efficiency of CNN-based AI models, such as ToothNet and SSD MobileNetV2, was evaluated against dental specialists in two studies[18,38]. ToothNet has reported a higher sensitivity (91.1%) than the dentist (63.7%.) However, SSD MobileNetV2 had a sensitivity that ranged from 0.0 to 0.76, but higher performance was found for the dentist having a sensitivity range of 0.77 to 1.0.

Explainability

Out of 23 studies, 3 employed XAI to visualize the segments of an image that were highly significant for the model's reasoning[18,26,31]. Additionally, a level 2 deployment status was achieved by 17 studies that assessed the AI models and tested their efficacy on dataset photos without any comparisons or real-world application[13,19,23–28,30,31,33–37,39,40].

Level of deployment

Level 3 deployment status was achieved by 1 study which compared the model (ToothNet) with dental experts possessing 1 year of experience[18]. A total of 4 studies exhibited a level 4 deployment status by developing mobile or web-based applications accessible to the general population[17,32,38,41].

DISCUSSION

AI has been harnessed in different aspects of dentistry, including dental caries diagnosis. Detection of caries, especially during its early phase, is of paramount importance to mitigate future sequelae in terms of health and treatment costs[42]. AI-based DL models have been utilized to detect dental caries using intraoral images[17]. Generally, it was noted that the performance of different

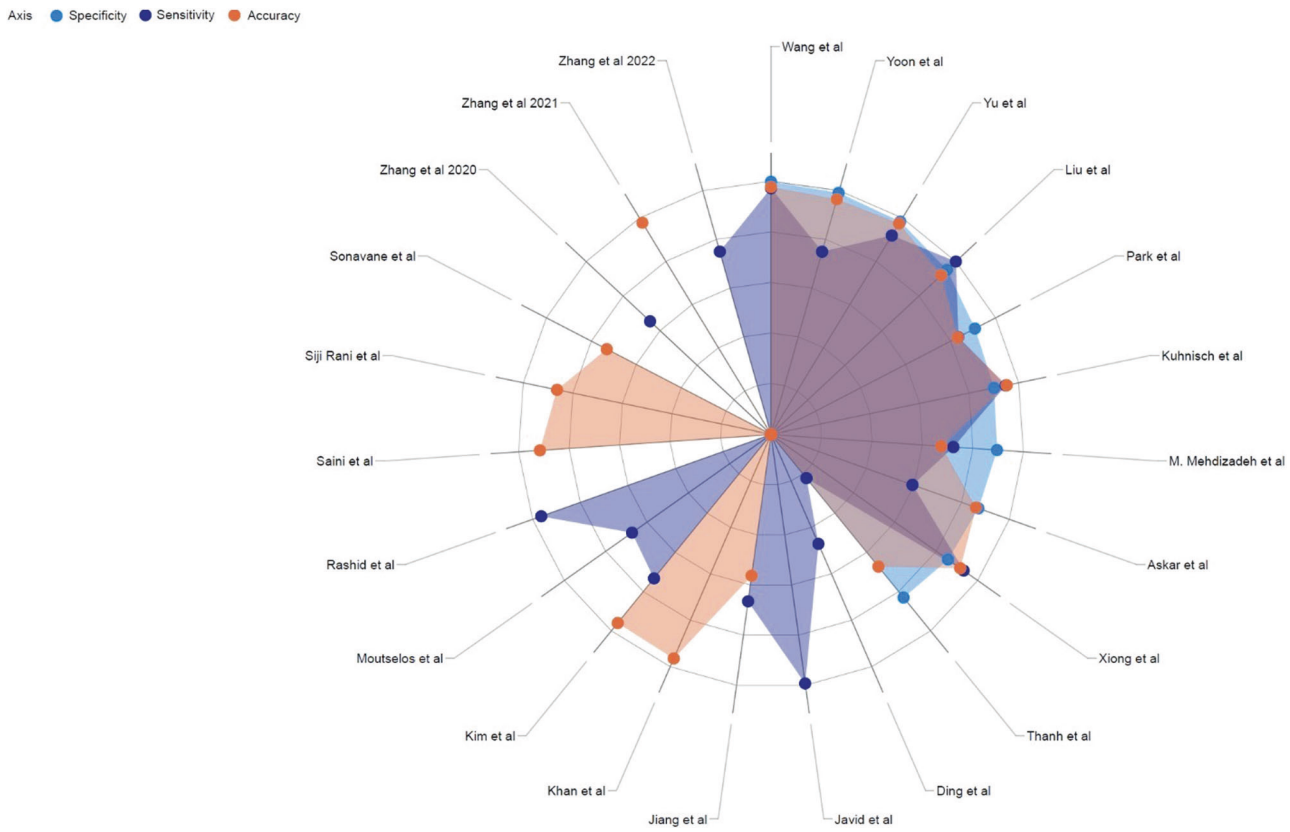


Fig. 3 Specificity, Sensitivity, and Accuracy of all the included studies. (n=23) The radar chart illustrates the specificity, sensitivity, and accuracy metrics from various studies. Each axis represents a different study, with the values of specificity, sensitivity, and accuracy plotted radially from the center. The outermost circle denotes a value of 100%, and each subsequent inner circle represents a 20% decrement, with the center point indicating a value of 0%.

AI models varied significantly. Nevertheless, AI models are promising tools to be used for the detection of dental caries.

The studies included in this review relied on multicenter datasets. The differences in demographics and data quality across various centers introduces variability making it possible to generalize the findings to broader populations. Furthermore, it was observed that most of the studies were primarily efficacy studies, focusing on how well the interventions perform under controlled conditions. In contrast, only a small number of studies assessed effectiveness, which evaluates how well these interventions work in real-world settings.

Quality assessment of the included studies was performed using the Modified-QUADAS 2 tool. Few studies included in this systematic review had a high risk of bias, which is in line with previous reviews[17,23–25]. The performance of AI-models is primarily judged on the basis of accuracy, sensitivity, and specificity metrics. These performance metrics were reported to be greater than 80%, corresponding to the previous review[16].

AI has been utilized globally, however, disparities are seen within the high-income and low-income countries. This review found that most of the studies were conducted in high-income countries. Greater utilization of resources has been observed in high-income countries, this could be due to reasons such as support from institutions and policymakers, ample funding, as well as increased awareness. On the other hand, low-income countries are mainly deprived of such resources. Hence, a large gap is observed in terms of utilization of advanced technologies such as AI. To bridge this gap, collaborative research networks and multicenter studies need to be encouraged. This would enable the sharing of resources, expertize, and infrastructure, accelerating AI adoption in underserved regions.

Some studies in this review acquired data for AI-model training from online repositories such as Kaggle©, GitHub©, and Zenodo©. However, acquiring data from these sources comes with some limitations. Datasets available online are uploaded by users without checking their accuracy and reliability, hence they may lack generalizability[37]. Importantly, not all studies clearly mentioned the sources of data retrieval, making it difficult to verify the origin and quality of the data. It was also noted that two studies have used the same datasets for their model training and testing which may lead to redundancy and wastage of important resources[24,29]. Hence, it is recommended for further studies to acquire real time data from their own study settings so as to improve uniqueness.

For adequate performance of AI models, the training dataset should be balanced with sufficient examples of diseased as well as non-diseased teeth. This review found that some studies reported the prevalence of carious lesions in their dataset whilst others did not. For transparent and unbiased model training, it is crucial to specify the disease prevalence. Additionally, only a few studies have used classification systems such as ICDAS and ADA CCS. This lack of standardized systems affects the validity of inter-study comparisons. Since the results are not based on a uniform standard, it is harder to draw conclusions or generalize outcomes across studies.

Only two studies utilized visual-tactile examinations as ground truth to ensure the accurate diagnosis of carious lesions[28,37]. Different diagnostic techniques have varying thresholds, and most studies relied on visual annotations, which can introduce subjectivity and potential biases[26,27,30,31,33,34,36,38,39]. These biases may make AI-models to sometimes misinterpret stains as carious lesions. Therefore, future studies must establish more reliable and standardized diagnostic methods to avoid such false positives. Future studies should also describe their training methods and criteria, validation techniques, and calibration processes for better transparency. Additionally, explaining the processes for resolving disagreements of caries detection should be reported.

AI models are available either as off-the-shelf or custom models. Most of the studies in this review have employed the application of off-the-shelf AI models while some studies presented their custom CNN models such as UCDA, RDFNet, ToothNet, T-Net, and DentalNet[18,26,29,35,36]. Among these, the performance of UCDA and

RDFNet have overcome the performance of off-the-shelf AI models. Future studies should focus on developing custom AI models for improved performance outcomes.

In order to determine the performance of AI a head-to-head comparison with humans is critical. Such comparison of outcomes based on trained AI algorithms vs human experts ascertain its performance in real-life. In a few validation studies, dentists were assigned to the control group (primarily to establish as a reference test). In one study DL model DentalNet was found to be more accurate than humans, however, another study reported the latter to have a greater accuracy than the AI model[29]. These discrepancies suggest that AI model performance is highly context-dependent, influenced by factors such as the experience of human participants, the type and pre-processing of images and the picture-capturing device used. These variables highlight the need for more nuanced analysis in future validation studies to assess AI performance under varying conditions.

Despite the meticulous approach in conducting the included studies, some limitations and strengths were encountered. To evaluate the impact of AI, understanding the levels of deployment is critical. The levels of deployment of AI varies from 1 to 4 as follows: Level 1 as proof of concept. Level 2 as validation studies, Level 3 as comparing humans with AI, and Level 4 as real-world applications[10]. This review did not have any study falling under level 1 however, most of the studies achieved level 2 while only few studies reached level 4. The limited number of studies reaching level 4 deployment could be due to several key barriers. A major challenge in integrating AI into clinical workflows is the need for larger, standardized datasets, which require more computational power—often unavailable in clinical settings. Ethical concerns and data privacy issues add to the difficulty. Creating collaborative data frameworks would help ensure diverse datasets are available to support AI development and application in healthcare.

Conducting more level 4 studies may assess the performance of AI models in diverse populations and clinical settings, hence, improving the overall diagnostic accuracy. A study successfully developed a tool for dental caries detection which was rated as satisfactory/A+ by both dental experts and the general audience according to the System Usability Scale (SUS)[41].

Most studies have focused on the level 2 deployment status, where trained models were evaluated, lacking real-world applications. However, one study compared the model's performance with that of a dentist with only one year of experience achieving level 3 deployment status[18]. While the model performed better, it emphasizes the need to compare models against dental professionals with more years of experience.

AI is usually referred to as a 'black box' meaning that the internal decision-making processes are not easily understandable to humans[43]. XAI refers to the ability of AI algorithms to provide clear and understandable reasons for their decisions and recommendations[44]. It comprises of various algorithms and techniques that aim to make DL models more interpretable, explainable, and transparent. XAI techniques can help elucidate the decision-making processes of AI models used for caries detection. Few studies have applied XAI techniques such as Grad-CAM to interpret their model's predictions[26]. Consequently, most of the studies lacked transparent explanations of the generated outcomes. XAI also facilitates better clinical decision-making by providing explanations and the reasons behind their predictions —such as which specific features in the intraoral images indicate caries so that the dentists can make informed decisions and maintain the reliability of the AI model.

Regarding the strengths of this review, firstly, we included studies that employed XAI techniques for caries detection highlighting the transparency and interpretability of AI-models, essential for clinical acceptance. Secondly, the level of deployments of included studies were also evaluated whereby 4 studies reached level 4 deployment, hence indicating the practical application of AI-models for caries detection.

However, there were some limitations in this review. Only three databases were searched limiting the retrieval of potential studies. Initially, meta-analysis was planned for the outcomes of performance metrics. However, due to significant clinical and methodological heterogeneity, such quantitative data analysis was not possible. Additionally, excluding studies that reported non-DL AI models might have been a notable limitation. This may have introduced potential bias, as traditional ML models can also provide valuable insights. Recent studies have reported that ML algorithms generally have lower accuracy compared to DL models. For instance, one study found that a SVM achieved an accuracy of 78%[45], while another study reported that a CNN reached an accuracy of 97.7%[32]. This disparity highlights the importance of considering both ML and DL approaches in order to provide a more comprehensive understanding of their performance in caries detection.

CONCLUSION

The diagnostic performance of AI-models was found to vary across different AI-models. Generally, higher accuracy, sensitivity, and specificity were reported. Although some studies managed to reach the level 4 deployment it is recommended for such studies to get their real-world applications assessed. XAI has been reported to enhance the clinical decision-making process which improves the reliability of the AI-model. Hence, there is a need for future studies to explore the integration of explainability and assess its impact on decision-making accuracy.

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A.N: Writing – Original Draft, Data Collection, Data Analysis; **S.A.A:** Writing – Original Draft, Data Collection, Data Analysis**, A.L:** Writing – Original Draft, Data Analysis, **N.A:** Writing – Original Draft, Data Collection, Data Analysis**, S.M.F.A:** Data Analysis**, F.U:** Conceptualization, Study Design, Writing – Original Draft, Data Analysis, Final Approval of Manuscript.

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